

Claim Severity Models and Actuarial Statistics

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September 30, 2026

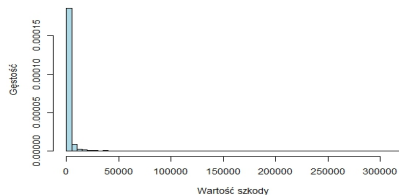
Database description

- **Dataset:** `frecomfire` (*CASdatasets* package).
- **Scope:** Commercial fires in France.
- **Variable:** `ClaimCost2007` – claim amount.
- **Characteristics:** Highly asymmetric, right-skewed distribution with a heavy tail.

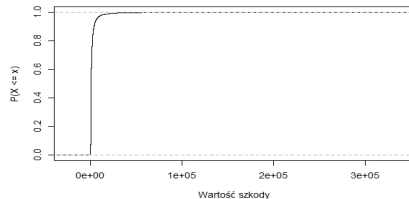
Data Analysis – Empirical Estimation

- Examination of the empirical distribution of the original and log-transformed variable.

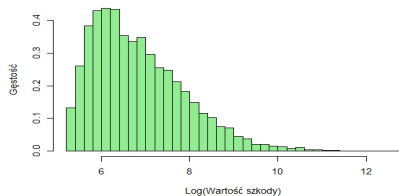
Histogram (Original)



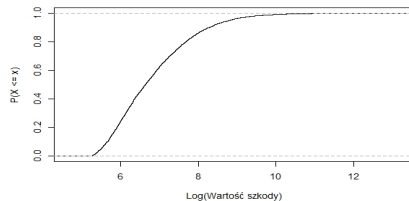
Dystrybuanta empiryczna (Original)



Histogram (Logarytm)

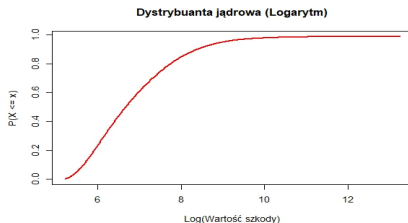
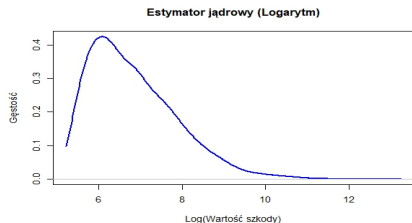
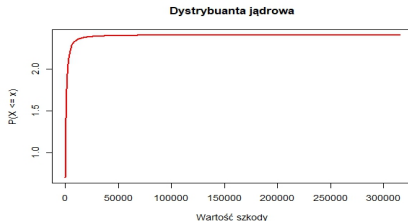
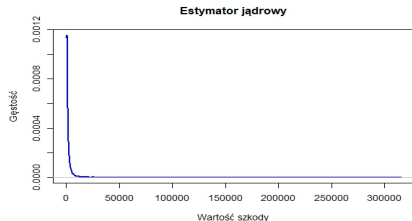


Dystrybuanta empiryczna (Logarytm)



Data Analysis – Kernel Estimation

- Non-parametric density smoothing using a kernel estimator.



Descriptive Statistics of Distributions

For kernel estimation, a smoothing parameter (bandwidth) of $bw = 100$ was assumed for original data and $bw = 0.2$ for log-transformed data.

Table: Comparison of descriptive statistics for original and log-transformed data

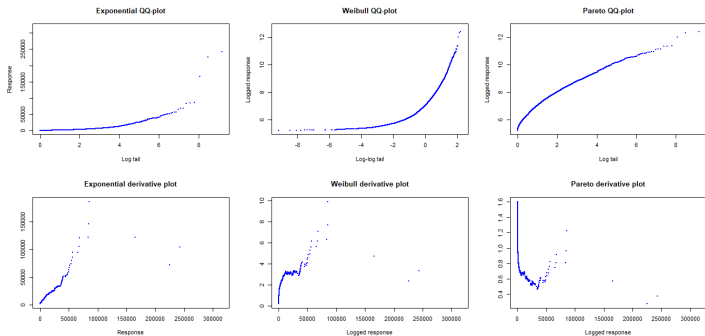
Statistic	Empirical (Orig.)	Kernel (Orig.)	Empirical (Log)	Kernel (Log)
Expected value	1981.124	1981.124	6.829852	6.829852
Standard deviation	6463.098	6463.871	1.041219	1.060254
Quantile 0.75	1750.547	1748.623	7.467684	7.472780
Quantile 0.95	6621.264	6637.185	8.798041	8.809994
Quantile 0.99	19877.987	20005.972	9.897368	9.934023

- Smoothing with a Gaussian kernel causes a slight increase in standard deviation and extreme quantile values (0.95 and 0.99) in the distribution tail.

Distribution Tail Analysis

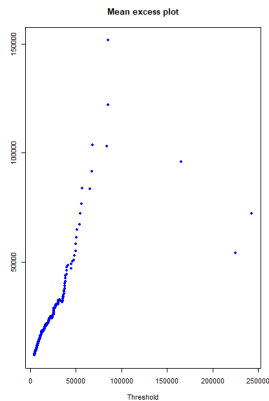
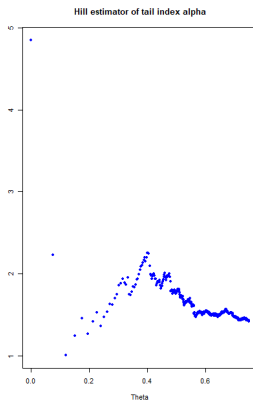
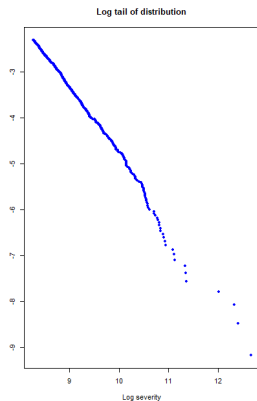
The empirical tail behavior is best described by the **Pareto distribution**:

- **Q-Q Plot:** The curve for the Pareto distribution deviates the least from the $y = x$ line.
- **Derivative Plot:** The derivative plot for the Pareto model is the most stable.



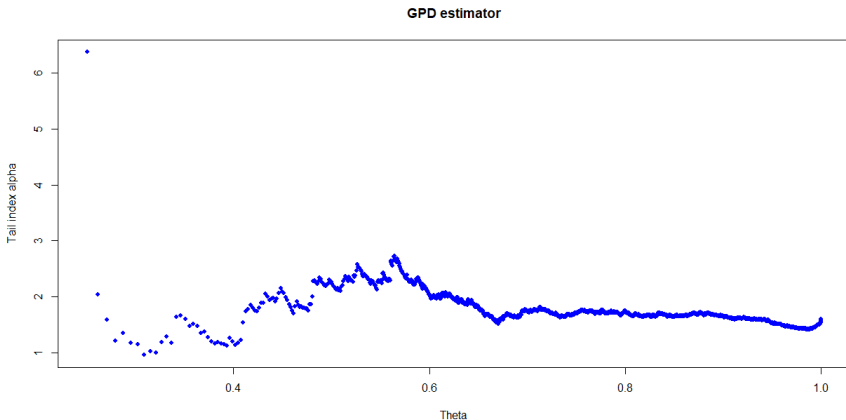
Tail Parameter Estimation – Pareto Distribution

- The analysis was conducted for a threshold at the 0.9 quantile.
- However, the mean excess plot indicated a cutoff threshold of $u = 40\,000$ (approx. 99.75th percentile).
- The Hill estimator indicates a stable parameter value at $\hat{\theta} \approx 0.58$.



Tail Parameter Estimation – GPD

- Based on the GPD estimator, the parameter value stabilizes around $\hat{\theta} \approx 0.65$.



Pareto Distribution – Characteristics and Goodness-of-Fit Tests

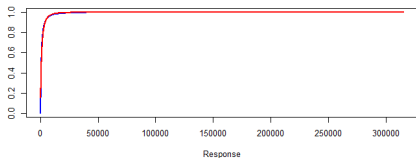
Measure / Quantile	Data	Model
Mean	1981.12	1802.29
Std. dev.	6463.10	3937.74
Skewness	25.56	0.00
Kurtosis	988.41	0.00
Q 70%	1426.81	1680.50
Q 80%	2156.18	2451.92
Q 90%	3907.30	4093.81
Q 95%	6636.55	6253.18
Q 99%	20147.99	14265.28

Statistical Tests

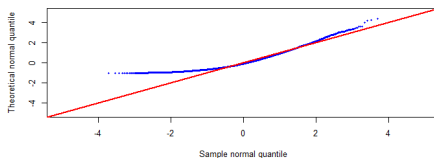
- **AIC Criterion:** 161291.8
- **Anderson-Darling Test:**
Statistic = 282.19
 $p\text{-value} \approx 0$ (Rejection of H_0)
- **Kolmogorov-Smirnov Test:**
 $p\text{-value} \approx 0$ (Rejection of H_0)

Pareto Distribution – CDF Visualization

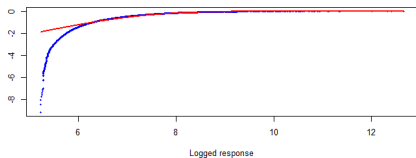
Distribution function



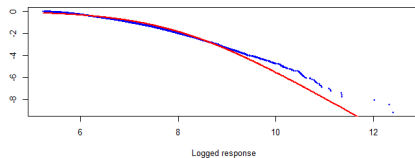
QQ plot



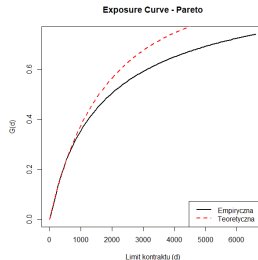
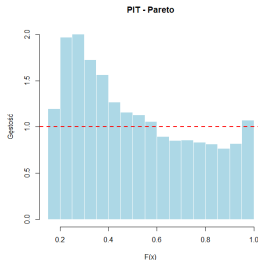
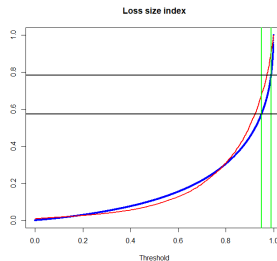
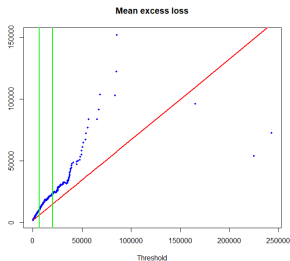
Logged distribution function



Logged survival function



Pareto Distribution – Diagnostic Plots



Pareto Distribution – Summary

- The model noticeably underestimates predictions for extremely high claims. The actual 99% quantile is over 20 147, while the model indicates only 14 265.
- The significant difference in standard deviation (data: 6463, model: 3937) proves that the classical Pareto distribution fails to capture the full variance of the sample.
- Visual analysis of the diagnostic plots indicates that the distribution fit is not fully satisfactory (which is consistent with the statistical test results, where p -value ≈ 0).
- The Pareto model shows great similarity to the actual data (especially in capturing the heavy tail), however, it does not perfectly reflect the structure of the entire distribution.

Gamma Distribution – Characteristics and Goodness-of-Fit Tests

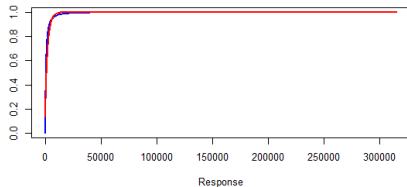
Measure / Quantile	Data	Model
Mean	1981.12	1980.99
Std. dev.	6463.10	2240.75
Skewness	25.56	2.26
Kurtosis	988.41	10.68
Q 70%	1426.81	2328.90
Q 80%	2156.18	3241.31
Q 90%	3907.30	4844.00
Q 95%	6636.55	6480.21
Q 99%	20147.99	10351.31

Statistical Tests

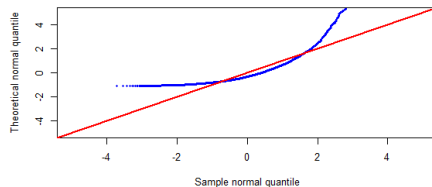
- **AIC Criterion:** 164761.19
(Log-Likelihood: -82378.59)
- **Anderson-Darling Test:**
Statistic = 555.72
 p -value ≈ 0 (Rejection of H_0)
- **Kolmogorov-Smirnov Test:**
 p -value ≈ 0 (Rejection of H_0)

Gamma Distribution – CDF Visualization

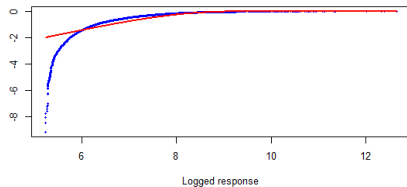
Distribution function



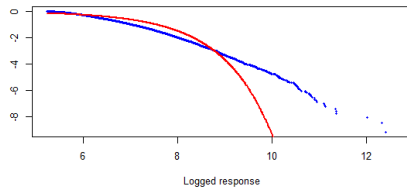
QQ plot



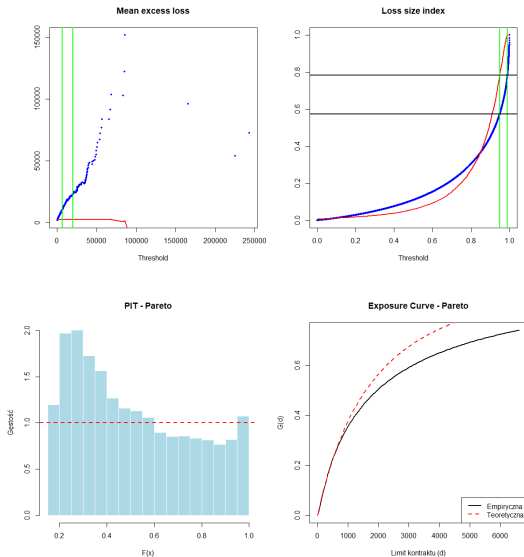
Logged distribution function



Logged survival function



Gamma Distribution – Diagnostic Plots



Gamma Distribution – Summary

- The Gamma distribution well reproduces the mean claim amount (data: 1981.12, model: 1980.99). Unfortunately, its light tail causes a drastic underestimation of extremes (Q 99% underestimated by half).
- Significantly underestimated standard deviation (2240 vs 6463) and much lower kurtosis suggest that the model does not generate rare claims.
- Visual analysis of the diagnostic plots shows poor model fit to the empirical data.
- This distribution belongs to the class of light-tailed distributions, thus its fit to the analyzed data is incorrect and does not allow for a proper representation of the claim structure.

Fréchet Distribution – Characteristics and Goodness-of-Fit Tests

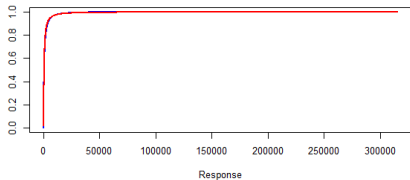
Measure / Quantile	Data	Model
Mean	1981.12	2622.44
Std. dev.	6463.10	∞
Skewness	25.56	∞
Kurtosis	988.41	∞
Q 70%	1426.81	1311.76
Q 80%	2156.18	1907.61
Q 90%	3907.30	3473.03
Q 95%	6636.55	6170.44
Q 99%	20147.99	22673.59

Statistical Tests

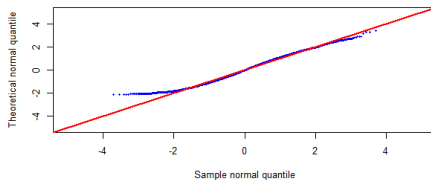
- **AIC Criterion:** 157623.79
(Log-Likelihood: -78809.89)
- **Anderson-Darling Test:**
Statistic = 28.32
 p -value ≈ 0 (Rejection of H_0)
- **Kolmogorov-Smirnov Test:**
 p -value ≈ 0 (Rejection of H_0)

Fréchet Distribution – CDF Visualization

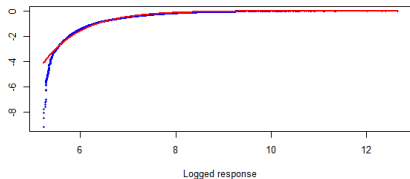
Distribution function



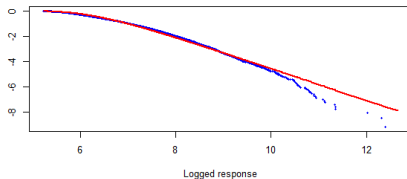
QQ plot



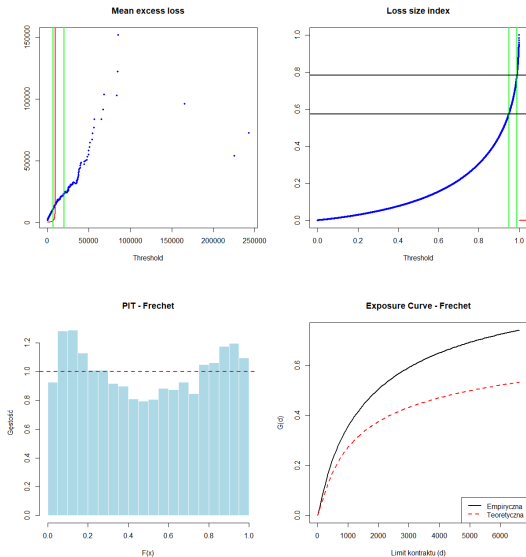
Logged distribution function



Logged survival function



Fréchet Distribution – Diagnostic Plots



Fréchet Distribution – Summary

- The model exhibits infinite variance and higher moments. This is due to the very heavy tail of the distribution, which is visible in the 99% quantile, slightly overestimated (22 673 compared to 20 148).
- The AIC criterion (157 623) is the lowest among the models examined so far. The Anderson-Darling statistic dropped to 28.32 (from 555 for Gamma and 282 for Pareto), indicating a better fit in the right tail.
- Despite visible improvement, formal A-D and K-S tests still reject the null hypothesis (p -value ≈ 0), which is typical for large datasets with complex structures.
- Plot analysis confirms that the Fréchet distribution well describes the largest claims. However, attention should be paid to the mean discrepancy (data: 1 981.12, model: 2 622.22).
- Unlike other analyzed models, the theoretical exposure curve for the Fréchet distribution runs below the empirical curve, which means a systematic underestimation of total claim amounts and insufficient representation of tail heaviness.

Loglogistic Distribution – Characteristics and Goodness-of-Fit Tests

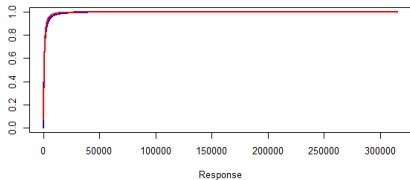
Statistical Tests

Measure / Quantile	Data	Model
Mean	1981.12	1605.85
Std. dev.	6463.10	∞
Skewness	25.56	∞
Kurtosis	988.41	∞
Q 70%	1426.81	1382.65
Q 80%	2156.18	1895.57
Q 90%	3907.30	3047.18
Q 95%	6636.55	4719.10
Q 99%	20147.99	12402.39

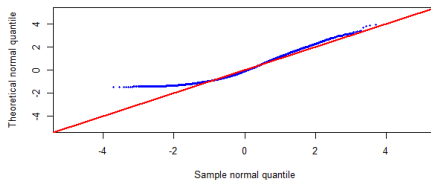
- **AIC Criterion:** 159305.56
(Log-Likelihood: -79650.78)
- **Anderson-Darling Test:**
Statistic = 104.85
Rejection of H_0 at the level $\alpha = 0.05$
- **Kolmogorov-Smirnov Test:**
Rejection of H_0 at the level $\alpha = 0.05$

Loglogistic Distribution – CDF Visualization

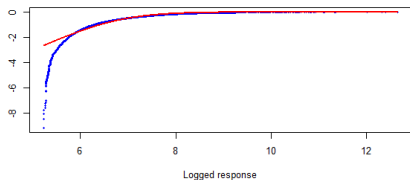
Distribution function



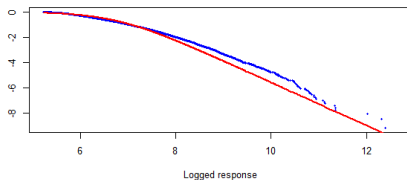
QQ plot



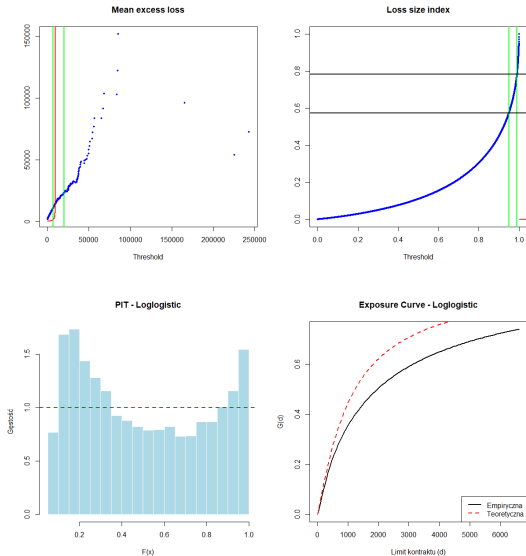
Logged distribution function



Logged survival function



Loglogistic Distribution – Diagnostic Plots



Loglogistic Distribution – Summary

- The loglogistic distribution shows a stable fit outside the tail and moderately for extreme claim values. The Anderson-Darling statistic result (104.85) confirms a good quality of data structure description, although this is a slightly weaker result than those obtained for the Fréchet distribution.
- Despite theoretical heavy-tailedness, the model tends to underestimate extreme events. The 99% quantile value is 12 402 compared to the actual value of 20 148.
- The Akaike information criterion is (159 305), positioning this model as more precise than the Gamma and Pareto distributions, but ultimately it is a slightly worse fit than the preferred Fréchet distribution (157 623).
- Analysis shows that the loglogistic distribution is slightly inferior in terms of tail flexibility, making it a second-choice model compared to the Fréchet model.

Spliced Model (Fréchet-Loglogistic) – Characteristics and Tests

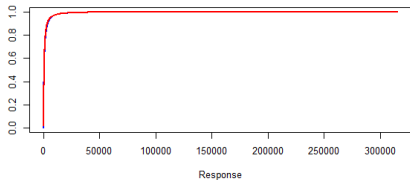
Measure / Quantile	Data	Model
Mean	1981.12	1919.38
Std. dev.	6463.10	∞
Skewness	25.56	∞
Kurtosis	988.41	∞
Q 70%	1426.81	1322.71
Q 80%	2156.18	1924.07
Q 90%	3907.30	3487.01
Q 95%	6636.55	6101.20
Q 99%	20147.99	19498.79

Statistical Tests

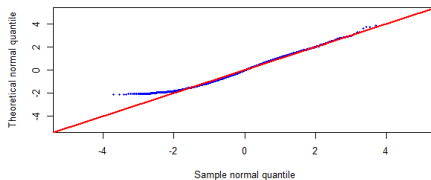
- **AIC Criterion:** 157608.09
(Log-Likelihood: -78800.04)
- **Anderson-Darling Test:**
Statistic = 27.62
 p -value ≈ 0 (Rejection of H_0)
- **Kolmogorov-Smirnov Test:**
 p -value ≈ 0 (Rejection of H_0)

Spliced Model (Fréchet-Loglogistic) – CDF Visualization

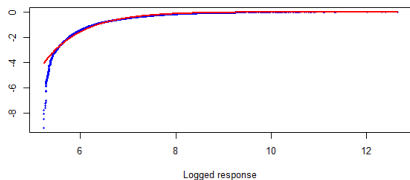
Distribution function



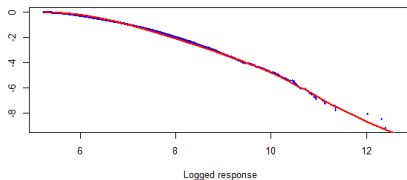
QQ plot



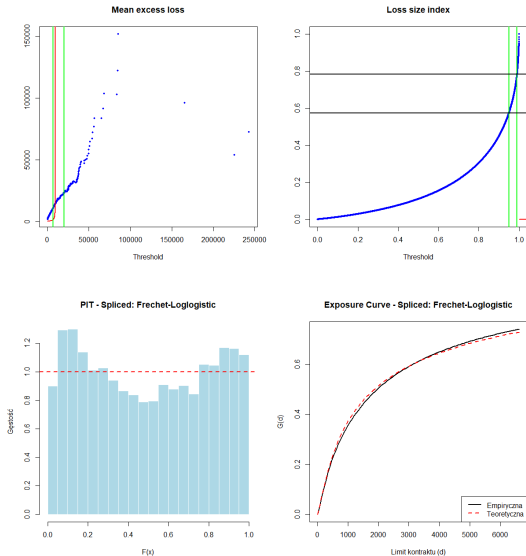
Logged distribution function



Logged survival function



Spliced Model (Fréchet-Loglogistic) – Diagnostic Plots



Spliced Model (Fréchet-Loglogistic) – Summary

- The spliced model achieved the lowest (best) AIC criterion value so far (157 608.09), surpassing the independently analyzed Fréchet distribution. This indicates that the use of spliced distributions was justified.
- The Anderson-Darling statistic reached the lowest level (27.62). The 99% quantile (19 498.79) very accurately approximates the empirical extreme value (20 147.99), eliminating the problem of excessive overestimation visible in the Fréchet distribution alone.
- Applying the Fréchet distribution for claims below the threshold allowed maintaining the model mean (1919.38) within an acceptable proximity to the empirical mean (1981.12).
- The structure combining the Fréchet and loglogistic distributions in the tail is a better alternative to the Fréchet distribution, due to slightly better diagnostic results, particularly a mean close to the empirical data.

Spliced Model (Fréchet-Pareto) – Characteristics and Tests

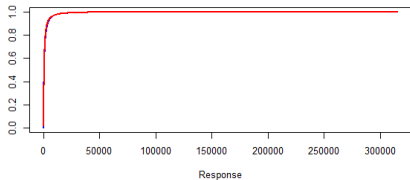
Measure / Quantile	Data	Model
Mean	1981.12	1916.86
Std. dev.	6463.10	∞
Skewness	25.56	∞
Kurtosis	988.41	∞
Q 70%	1426.81	1322.71
Q 80%	2156.18	1924.07
Q 90%	3907.30	3487.01
Q 95%	6636.55	6101.20
Q 99%	20147.99	19498.79

Statistical Tests

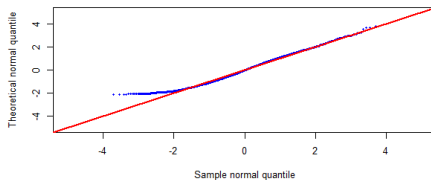
- **AIC Criterion:** 157609.8
(Log-Likelihood: -78800.9)
- **Anderson-Darling Test:**
Statistic = 27.62
 p -value ≈ 0 (Rejection of H_0)
- **Kolmogorov-Smirnov Test:**
 p -value ≈ 0 (Rejection of H_0)

Spliced Model (Fréchet-Pareto) – CDF Visualization

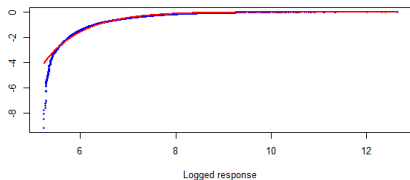
Distribution function



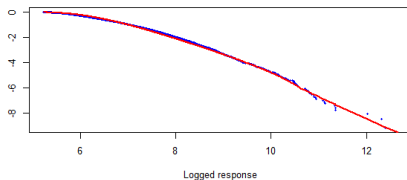
QQ plot



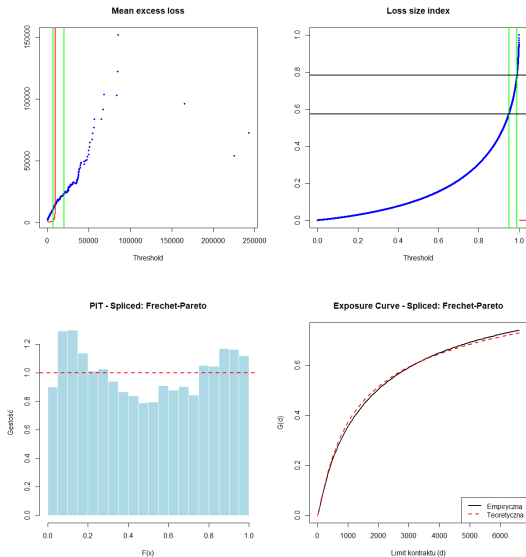
Logged distribution function



Logged survival function



Spliced Model (Fréchet-Pareto) – Diagnostic Plots



Spliced Model (Fréchet-Pareto) – Summary

- The Fréchet-Pareto and Fréchet-Loglogistic distributions are characterized by an almost identical fit structure to empirical data, providing very high quality in representing claims, especially for extreme values.
- Diagnostic values for both models are similar – the Anderson-Darling statistic at 27.62 and marginal differences in the AIC criterion confirm that in actuarial practice, both approaches provide practically the same results.
- The choice of the Fréchet-Pareto model was primarily dictated by practical considerations and ease of implementation, resulting from the use of the classical Pareto distribution.

EM Algorithm (Exponential Distribution) – Characteristics and Tests

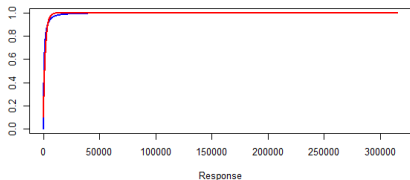
Statistical Tests

Measure / Quantile	Data	Model
Mean	1981.12	1767.17
Std. dev.	6463.10	1767.17
Skewness	25.56	2.00
Kurtosis	988.41	9.00
Q 70%	1426.81	2127.63
Q 80%	2156.18	2844.15
Q 90%	3907.30	4069.06
Q 95%	6636.55	5293.97
Q 99%	20147.99	8138.12

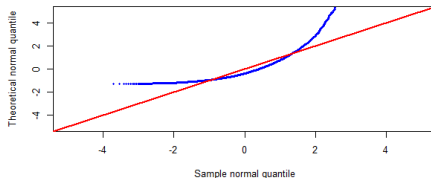
- **AIC Criterion:** 165311.12
(Log-Likelihood: -82654.56)
- **Anderson-Darling Test:**
Statistic = 530.84
 p -value ≈ 0 (Rejection of H_0)
- **Kolmogorov-Smirnov Test:**
 p -value ≈ 0 (Rejection of H_0)

EM Algorithm (Exponential Distribution) – CDF Visualization

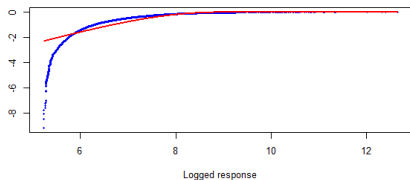
Distribution function



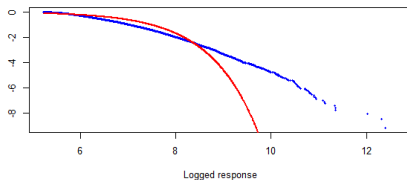
QQ plot



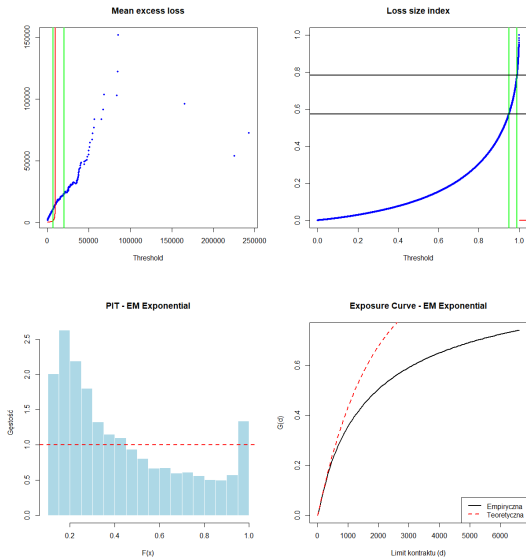
Logged distribution function



Logged survival function



EM Algorithm (Exponential Distribution) – Diagnostic Plots



EM Algorithm (Exponential Distribution) – Results and Modifications

- Modification of the Expectation-Maximization (EM) algorithm consisted of analytically deriving the objective function in the Expectation step, thereby replacing numerical approximations.
- Distribution rigidity (the requirement to equate the mean with the standard deviation, 1767.17) means that the model does not represent the high dispersion of the empirical data (6463.10).
- The light tail of the distribution results in underestimating extreme risk – the theoretical 99% quantile is only 8138 compared to the actual 20 147.
- The weakest goodness-of-fit indicators (AIC: 165 311, A-D statistic: 530.84) confirm that despite the correct implementation of EM, this distribution is insufficient. This justifies the necessity of using spliced models, such as Fréchet-Pareto.

Point estimates – Method comparison

Risk Measure	Spliced Model	GPD Approximation
$Var_{0.995}$	29 184.53	30 715.35
$ES_{0.995}$	64 744.88	60 936.98

Confidence intervals (t-bootstrap) for the spliced model

Bootstrap Method	Interval for $Var_{0.995}$	Interval for $ES_{0.995}$
Non-parametric	[19561.18, 31427.49]	[−2235.12, 80161.95]
Parametric	[25655.19, 31847.92]	[−67705.20, 84700.31]

Final conclusions

- Among the classical distributions, the Fréchet distribution showed the best fit, but its application was abandoned due to significant overestimation of the empirical mean. Spliced distributions provided the highest quality of data representation, from which the combined Fréchet-Pareto distribution was selected for the final analysis.
- Risk measure estimates obtained from the spliced model are highly convergent with the results of the independent GPD approximation. The Value-at-Risk measure itself shows high stability, and compact confidence intervals allow for reliable determination of capital requirements.
- The bootstrap method generated confidence intervals encompassing negative values for the Expected Shortfall, proving the massive sensitivity of this measure to single, catastrophic claims in the historical sample.
- This phenomenon is a consequence of an extremely heavy tail. For the shape parameter $\xi = 0.65 > 0.5$, the theoretical variance of the distribution tends to infinity.